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Enhancing User Experience: Immersive Virtual Reality Property Showhouse

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***Abstract*— This paper presents the design and development of an immersive Virtual Reality (VR) Property Showhouse application and an initial study that explores the use of VR technology in enhancing user experience for viewing property showhouse. Despite virtual tour facilitating the visualization and marketing of actual properties via digital platforms, the persisting inadequacy of conventional approaches in delivering an authentic and immersive portrayal of property show units remains a challenge. In pursuit of addressing this challenge, the immersive VR Property Showhouse was developed to empower users to delve into Digital Home virtual tour in a more experiential and engaging manner. By capitalizing on immersive VR technology, users can traverse properties in a highly captivating and verisimilar fashion, cultivating a sense of presence and simulating the sensation of physically inhabiting the space. The technical details which include the design considerations for creating the immersive user experience for the app are presented. This paper additionally examines closely the existing literature on VR's application in real estate, showcasing the prospective advantages of this technological marvel. To gauge the efficacy of the VR application a usability evaluation was conducted using the System Usability Scale (SUS) with 30 participants. The outcome of the user evaluation manifested a positive user experience, with an average SUS score of 85, signifying exceptional usability. These results provide important knowledge for those looking to integrate immersive VR into the real estate industry and contribute to understanding how VR can be designed and used effectively to enhance user engagement and understanding of properties, ultimately influencing their decision-making process. Given the soaring popularity of virtual tours, it is important for real estate practitioners and VR developers to continuous seek innovative approaches to refine the features and immersion quality of these tours, ultimately delivering an unparalleled experience for potential buyers.**

Keywords—Immersive Virtual Reality, Property Showhouse, User Experience Enhancement, Virtual Tours, Real Estate Industry

I. INTRODUCTION

The process of showcasing properties to prospective buyers has experienced a profound change in recent years, driven by notable advancements in technology and the ever-evolving needs of discerning consumers. Traditionally, property viewing heavily relied upon physical visits, wherein potential buyers embarked on personal explorations, harnessing their senses and imagination to envision their future living spaces. However, the advent of technology has ushered in alternative modalities, including websites, AR, and non-immersive VR, providing new ways to experience viewing of properties remotely. Nevertheless, the emergence of immersive VR has engendered a revolutionary paradigm shift, fundamentally altering the landscape of property showcasing and experiential engagement.

Following the devastating COVID-19 pandemic, the limitations of physical visits have been revealed, emphasizing the critical need for remote alternatives to address the current circumstances and challenges. By assuming a pivotal role, technology has effectively bridged the gap, granting prospective buyers the empowering ability to remotely view properties from the secure confines of their own homes, ensuring their safety and convenience while deciding on their dream house. Websites have emerged as robust platforms facilitating the perusal of property listings and dispensing crucial information, thereby endowing convenience and broadening the horizons of property exposure. However, their limitations lie in their inability to fully immerse users in the property's environment, leaving room for uncertainty and limited visualization. In contrast, immersive VR, with its distinct potentials, ascends as a transformative solution, augmenting the property viewing experience to unprecedented heights. Through the utilization of cutting-edge VR technologies, prospective buyers can now partake in immersive, interactive, and remarkably realistic virtual property tours that foster a profound sense of presence, replicating physical experiences seamlessly and, in turn, enhancing comprehension and facilitating the decision-making process with heightened efficacy.

While the potential benefits of immersive VR in the context of property showcasing are obvious, it is critical to acknowledge the existence of research gaps that need to be explored and elucidated further. One such gap pertains to user acceptance, particularly among the younger generation, who are more accustomed to digital experiences and technology-driven interactions. Determining the perception and acceptance of employing immersive VR for property viewing is important to propel the adoption of this technology. Consequently, this research work describes the design and implementations of an immersive VR Property Showhouse that offers potential buyers an engaging and informative experience, with the objective of enriching their understanding of the property. The VR Property Showhouse presents an immersive virtual environment that enables prospective buyers to navigate through the property, affording them a remarkably realistic experience akin to an in-person visitation. The VR Property Showhouse provides visitors with a seamless experience to explore the property's various rooms. This enhances understanding of spatial layouts and captures the distinct ambiance of the premises. Besides, the VR application empowering visitors to tailor their viewing experience to their individual preferences. For example, visitors can modify the interior wall colours enabling them to envision an ideal living space. More importantly, the VR Property Showhouse transcends geographical boundaries, letting potential buyers to remotely explore the property from any location. This feature confers distinct advantages, particularly for international buyers or individuals who are physically unable to undertake a personal visitation. The purpose of this research work was to examine the user experience and acceptance of the immersive VR Property Showhouse application for the younger generation. A sample of 30 young participants, ranging in age from 18 to 24, from Malaysia, took part in the user evaluation. The collected data was subjected to a quantitative analysis employing the System Usability Scale (SUS) questionnaire. By delving into the participants' experiences, attitudes, and overall satisfaction with the VR Property Showhouse application's features, usability, and user satisfaction, this research aimed to contribute to a deeper understanding of the existing knowledge of using immersive VR for property showcasing. The investigation offers a fresh perspective on the potential advantages and limitations of immersive VR in the property showcase domain, thereby offering valuable insights for developers, real estate professionals, and researchers striving to enhance the property viewing experience. The remaining manuscript proceeds as follows: Section 2 provides a brief overview of the utilization of different VR forms in the real estate sector. Section 3 provides comprehensive details on the design and implementation of the VR Property Showhouse. Section 4 discusses the evaluation methodology for this study and presents the findings of the research, focusing on the usability of the application. The final section concludes the findings and proposes directions for future research.

II. Related Work

This section provides background information on the existing adoption of VR technologies by researchers and the real estate sector for virtual property tours. A great deal of previous research into virtual property tours has focused on utilizing web-based VR, desktop-based VR, smartphone-based VR, and headset-based VR to provide potential buyers with alternative ways of viewing properties, with discussion on the benefits associated with each approach.

Amidst the formidable COVID-19 pandemic, the real estate industry has seen a surge in the incorporation of a myriad of modern technologies including VR technologies for virtual tours [1]. The use of innovative and disruptive technologies in the real estate sector has grown in importance as they provide new avenues for innovation and business development [2]. Non-immersive web-based VR has emerged as a popular method of VR adoption, garnering significant traction and appeal among users [3]. This is because accessing virtual tours and listings through a website is very easy and can be done on computers and mobile devices. A study conducted by the National Association of Realtors revealed that, in 2022, 41% of homebuyers utilized online platforms to search for real estate [4]. The utilization of web-based VR in the real estate industry has had a considerable impact, augmenting the buying and selling process by providing potential buyers with a non-immersive and interactive experience. Consequently, this empowers them to make more informed decisions, thereby enhancing overall customer satisfaction [5]. Research studies have reported that web-based VR significantly improves the experiences of prospective buyers, facilitating their comprehension of the layout and features of pre-sale housing units [6]. Desktop-based VR applications are another popular type of VR technology that is utilized in the real estate sector to offer users a non-immersive experience. These applications leverage computers and monitors to provide users with a virtual experience of real estate properties. The findings of various studies indicate that this VR approach exerts a positive influence on customers' attitudes towards properties and service providers, leading to intensified levels of customer satisfaction and trust [7].

In order to provide a more captivating VR experiences, the real estate sector has delved into immersive VR technology. This includes the use of cost-effective smartphone-based VR applications that allow prospective buyers to participate in 360-degree virtual property tours [8]. These applications utilize smartphones with cardboard headsets, such as Google Cardboard and Samsung Gear, which furnish an economically viable means for virtual property exploration. The affordability and cross-compatibility of these cardboard headsets with diverse smartphones contribute to their accessibility and user-friendly nature. The adoption of immersive VR technology in the real estate sector provides new avenues for innovation and business development [9]. Nevertheless, it is imperative to acknowledge that this approach presents a beginner level of immersion, constrained head movements, and a shortage of advanced features like hand tracking and positional tracking. Moreover, the visual experience and responsiveness of the VR content may be influenced by the quality and capabilities of the user's smartphone, particularly with lower-end devices. To address these limitations, the real estate industry has has turned to headset-based VR applications that use Head-Mounted Displays (HMD) such as Meta Quest and HTC Vive. These applications aim to provide potential buyers with fully immersive, interactive, and captivating experiences during virtual property tours. Research has shown that this approach enhances the realism and immersion of property experiences, leading to improved decision-making [10]. Besides, reports indicate that headset-based VR applications have the potential to establish an emotional bond between users and properties, increasing the likelihood of a successful purchase [11]. Taken together, the previous studies support the notion of using VR technology for property showcasing to overcome the limitations of traditional property tours. As such, it is paramount to embark on the development of an immersive VR Property Showhouse that integrates top-tier visual elements, interactive functionalities, and an inherent feeling of presence, thereby bestowing upon potential buyers an unparalleled and enthralling experience that not only stimulates engagement but also facilitates decision-making throughout the property acquisition journey.

III. Design and Development of VR Property Showhouse

The VR Property Showhouse serves as an intricately crafted digital replica, mirroring an existing Digital Home located at the Faculty of Engineering, Multimedia University in Cyberjaya, Malaysia. Digital Home is a cutting-edge single-story residence, thoughtfully equipped with an array of smart devices, seamlessly integrating digital innovation into the realm of everyday living, thereby offering exceptional comfort and convenience. Crafted with precision, the immersive virtual environment has been purposefully developed to deliver an authentic and captivating experience to users embarking on the virtual tour. The development process involved various stages, encompassing intricate 3D modelling, diligent texturing, and careful implementation of the VR application, collectively contributed to the creation of a marvelous lifelike and immersive virtual representation of the Digital Home.

The development begins with an exhaustive crafting of a detailed floorplan of the physical Digital Home, as illustrated in Fig. 1. This comprehensive floorplan served as a vital foundation, accurately capturing crucial room dimensions, including wall heights and doorways, thus providing a robust basis for the subsequent 3D modelling process. The construction of the 3D models, intricately replicating walls, ceilings, and floors, commenced in an accurate adherence to the crafted floorplan, ensuring identical representation of the physical space. To accomplish the tasks of generating highly realistic and visually captivating assets, Maya, a renowned software for its proficiency in the field of 3D modelling, was employed. By harnessing the advanced capabilities and techniques of Maya, an array of 3D models was crafted and utilized to vividly depict the nuances of the Digital Home's furniture and environment, breathing life into the virtual space. The virtual furniture and home environment were sculpted with high accuracy by leveraging the discerning implementation of cutting-edge modelling techniques and incorporating the rich features offered by Maya, resulting in high quality of visual appeal, thereby enriching the immersive experience for users exploring the VR Property Showhouse. The augmentation of realism in the 3D models was achieved through the utilization of Substance Painter, an advanced software specifically designed for texturing. This proficient tool facilitated the creation of impeccably detailed textures and materials, including wood, fabric, and metal, which were applied to the virtual furniture and other elements within the immersive environment. The texturing process ensured that the virtual representations and their physical counterparts were highly similar, providing users with an immersive and visually stunning experience. Once all the home furniture were modelled and textured, they were imported into the captivating virtual environment through Unity3D for further development and coding. The selection of Unity3D as the software tool for constructing the immersive VR Property Showhouse was driven by its user-friendly interface and an extensive array of integrated functionalities. The virtual home environment provided an ideal starting point for constructing a cohesive property showcase experience. To enable a more immersive experience, advanced lighting techniques and an extensive repertoire of visual effects of Unity3D were utilized to enhance the virtual home environment.

To foster heightened engagement and interactivity within the VR Property Showhouse, an array of interactive features was designed and brought to life. For example, the ability to independently open and close doors, enabling visitors to embrace a spatial comprehension of the house's layout and a sense of realism. Besides, visitors can interact and examine various objects within the virtual home, facilitating

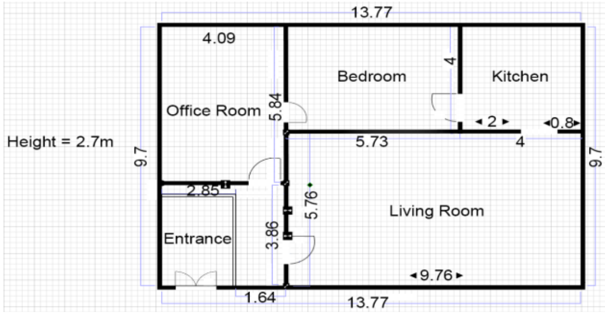

Fig. 1. Floorplan of Digital Home

engagement with the displayed items. All these implementations are done via Unity's powerful physics engines, which include collision detection mechanisms. For navigation and exploration freely within the VR Property Showhouse, the XR Interaction Toolkit was used to provide visitors with the option of using either teleportation or locomotion control methods. Furthermore, to enrich the informative aspect of the VR Property Showhouse, a 3D virtual assistant was developed and seamlessly integrated into the virtual home environment. This virtual assistant served as a guiding presence, offering essential information pertaining to the property, including its location, amenities, and floorplan. The detailed 3D model of the virtual assistant was created using Mixamo Fuse, while Mixamo was employed for rigging and animation, thereby enabling dynamic and responsive interactions with visitors, thus further enhancing the immersive nature of the virtual experience.

Upon donning an HMD and initiating the VR Property Showhouse, the demarcation between reality and virtuality becomes indistinct as visitors find themselves positioned at the entrance of the house, welcomed by a virtual salesperson eager to escort them through the Digital Home as depicted in Fig. 2. Assuming the role of a knowledgeable companion, the virtual salesperson accompanies visitors throughout their entire journey, augmenting their experience by imparting invaluable insights and facilitating a comprehensive comprehension of the showcased property. At the entrance, the virtual salesperson presents visitors with an interactive 3D floorplan, as illustrated in Fig. 3, enabling visitors to navigate the layout of the property, grasp spatial relationships, and acquire pertinent information pertaining to specifications and amenities. This empowers visitors to accurately evaluate the property's suitability for their distinct requirements and preferences. As visitors enter the VR Property Showhouse, a realm of boundless possibilities unfolds before their eyes. Seamlessly integrating intuitive

Fig. 2. Virtual Salesperson Guiding Property Exploration

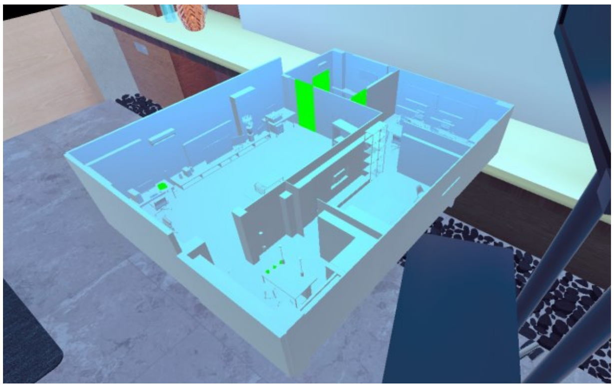

Fig. 3. Exploring Interactive 3D Floorplan

Fig. 4. Color Palette: Personalize the Virtual Space

controls and lifelike visuals, the application allows visitors to leisurely explore each area at their own pace, commencing from the entrance and extending to the living room, kitchen, bedroom, and office room. At each space, the virtual salesperson dispenses valuable insights, enabling visitors to understand the property's potential and fostering a vivid visualization of their ideal living space. The VR Property Showhouse provides a distinctive personalization feature to visitors through Color Palette as displayed in Fig. 4, in which a selection of colors can be used to alter the wall color within the virtual environment. With a few simple touches on the Color Palette, the walls come to life, transforming before the visitor's eyes. Visitors can instantaneously modify wall colors, thus enabling them to explore a multitude of aesthetic possibilities and discern the visual effects of different palettes on each room with ease and immediacy. The Color Palette lets visitors visualize their envisioned living environment, without necessitating physical alterations.

The immersive VR Property Showhouse Application redefines the paradigm for property exploration and assessment. It transcends the limitations of physical visits by providing an immersive experience that goes beyond what traditional methods can offer. The application enables visitors to effortlessly evaluate the property's suitability for their needs, offering a comprehensive understanding of its features and amenities. Guided by the virtual salesperson and unconstrained in their exploration of each space, visitors gain profound insights grounded in firsthand experiences and personalized interactions, thus fostering an enhanced grasp of the property's essence.

IV. USER EVALUATION

This section describes the methodology employed to conduct the pilot user evaluation of the immersive VR Property Showhouse application with the purpose of assessing the acceptance and usability among young adults aged between 18-24 years. The evaluation took place at the research Centre for Interactive Multimedia located at the Faculty of Creative Multimedia, Multimedia University that is equipped with PICO 4 VR headsets. A sample of 30 young adults within the target age group, with a mix of genders and varying levels of VR experience was selected to participate in the evaluation. Participants received no monetary or other compensation for their contributions.

Prior to initiating the actual evaluation, participants were asked to fill in a preliminary questionnaire to obtain demographic information, consent acquisition, and their familiarity with VR technology. Following this initial phase, a comprehensive demonstration session by a research assistant was given to provide detailed explanation to the functionalities of the VR Property Showhouse application. Subsequently, participants immersed themselves in an exercise, purposefully crafted to foster an acclimatization to the virtual realm before being assigned with actual tasks. The tasks in the evaluation were designed to assess the usability and functionality of the VR Property Showhouse which includes activities such as exploring different areas of the house, interacting with a virtual salesperson, modifying wall colors, and acquiring detailed information about the property. Upon the completion of these designated tasks, participants were directed to an online Google form, designed to obtain their feedback through an adapted version of the SUS questionnaire [12]. SUS questionnaire has been widely used across diverse domains, encompassing a broad range of products and user interfaces, including VR applications [13]. The questionnaire consists of 10 statements employing a user-friendly five-point Likert scale to obtain data. By analyzing the data captured through the evaluation, it can understand participants' perception of the VR Property Showhouse's usability, ease of use, learnability, and overall user satisfaction. After filling out the questionnaire, the participants were invited to take part in a post-evaluation interview that allowed them to share their qualitative reflections, introspective thoughts, and insightful observations. Through the interview, some participants provided additional comments, recommendations, and suggestions for future enhancement of the application.

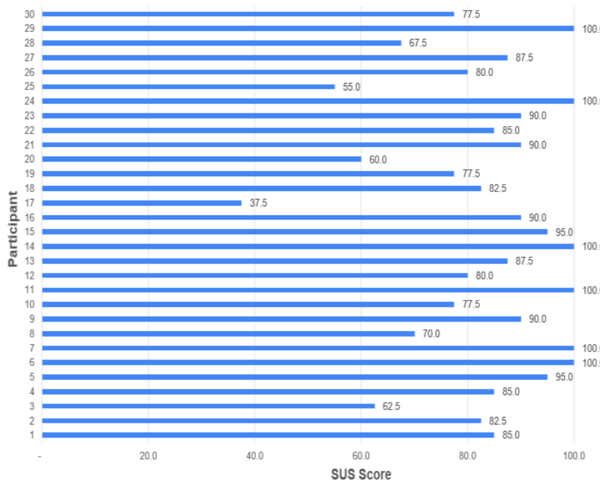

Fig. 5. SUS Scores of 30 Participants

The quantitative data obtained from the questionnaire were analyzed to determine the acceptance of the immersive VR Property Showhouse application. Fig. 5 provides a summary of the overall SUS scores, which revealed a high mean score of 83 and a median score of 85. The bar graph in also visually depicts the favorable level of usability observed among a majority of participants. These results suggest that the VR Property Showhouse provided young adult participants with an immersive and captivating user experience. Several participants expressed the feeling of physical presence within the showcased property, highlighting the application's effectiveness in delivering a convincing and immersive experience. The inclusion of a virtual salesperson significantly contributed to this sense of realism and excitement throughout participants' engagement with the virtual environment. Furthermore, the personalized feature of customizing wall colors enhanced engagement by enabling participants to explore various aesthetic possibilities and tailor the property according to their preferences. However, it is worth noting that participants' experiences exhibited variation, as indicated by a standard deviation (SD) of 15.08. While one participant reported a relatively low SUS score of 37.50, several others reported scores of 55.0 and 60.0, indicating the presence of difficulties encountered while using the VR Property Showhouse. Conversely, a significant majority of participants (20 out of 30) reported SUS scores of 80 or higher, suggesting that they were able to comprehend and utilize the application with ease. Furthermore, six participants gave the maximum SUS score of 100, underscoring the high potential of the VR Property Showhouse to be engaging and appealing to users.

Upon conducting an in-depth analysis of the individual SUS questions, several noteworthy findings have surfaced. Table 1 showcases the mean and median for each SUS question, providing valuable insights into the participants' perspectives. Result from the question 1 (M=4.27, Md= 4.00), indicates a strong positive inclination among the participants towards the VR Property Showhouse, perceiving it as appealing and engaging. Moreover, these scores imply a robust inclination among users to utilize the application regularly. As for Question 2 (M=1.67, Md=1.50), it shed light on the participants' perception of the application's complexity. Remarkably, the scores suggest that the majority of participants found the application to possess a user-friendly design, as it was not deemed unnecessarily complex. Notably, the guidance provided by the virtual salesperson, along with the informative dialogues, played a pivotal role in facilitating participants' understanding of the application's functionalities and features. The user interface and interaction design's effectiveness are evident from the participants' high scores for question 3 (M=4.43, Md=5.00), indicating their perception of the application as user-friendly, underscoring the seamless experience enabled by the application's design.

From score for question 4 (M=2.23, Md=2.00), it is suggested that while a portion of participants displayed a moderate inclination towards seeking guidance, the majority did not perceive a substantial reliance on external support when navigating through the application. Shifting focus to score of question 5 (M=4.33, Md=5.00), it illuminates participants' favorable perception of the application's capacity to facilitate seamless transitions between features, fluid examination of property particulars, and uninterrupted exploration of diverse spaces, culminating in a cohesive and harmonious user experience. Response of question 7 (M=4.33, Md=5.00) and question 8 (M=1.57, Md=1.00), suggesting that the application was easy to comprehend for the participants. They were able to swiftly engage with the VR experience and explore its features without encountering significant barriers or hindrances. Notably, the high score for Question 9 (M=4.57, Md=5.00) indicates participant were confidence and comfort in navigating the virtual property, exploring its features, and making informed decisions. This is further justified by the low score for Question 10 (M=1.73, Md=2.00), which implies that the participants did not perceive a significant learning curve when utilizing the application, suggesting an intuitive design that is easy to adopt. Analyzing the responses to question 7 (mean=4.33, median=5.00) and question 8 (mean=1.57, median=1.00) reveals that participants effortlessly grasped the application, swiftly immersing themselves in the VR experience and seamlessly navigating its features, thereby encountering minimal obstacles or impediments along the way. Remarkably, the high score attained for question 9 (mean=4.57, median=5.00) aptly signifies the participants' confidence and comfort as they traversed though the virtual property, exploring its myriad features, and judiciously arriving at well-informed decisions. The assertion is further reinforced by the relatively low score received for question 10 (mean=1.73, median=2.00), indicating the absence of a notable learning curve experienced by the participants during their interaction with the application, thus implying an intuitively designed application that seamlessly accommodates user adoption.

The evaluation results highlight the predominantly favorable experience of most young adults with the VR Property Showhouse, yet it is imperative to recognize the presence of variations in score, particularly in relation to question 4 concerning the requirement for guidance or support. The variations indicate that continuous improvement and refinement of the VR Property Showhouse are required to ensure an optimal user experience for all individuals.

TABLE I. RESULT FOR INDIVIDUAL SUS QUESTION

SUS Question		Mean (M)	Median (Md)
Q1	I think that I would like to use this Virtual Showhouse frequently.	4.27	4.00
Q2	I found this Virtual Showhouse unnecessarily complex.	1.67	1.50
Q3	I thought this Virtual Showhouse was easy to use.	4.43	5.00
Q4	I think that I would need assistance to be able to use this Virtual Showhouse.	2.23	2.00
Q5	I found the various functions in this Virtual Showhouse were well integrated.	4.33	5.00
Q6	I thought there was too much inconsistency in this Virtual Showhouse.	1.53	1.00
Q7	I would imagine that most people would learn to use this Virtual Showhouse very quickly.	4.33	5.00
Q8	I found this Virtual Showhouse very cumbersome/awkward to use.	1.57	1.00
Q9	I felt very confident using this Virtual Showhouse.	4.57	5.00
Q10	I needed to learn a lot of things before I could get going with this Virtual Showhouse.	1.73	2.00

V. CONCLUSION

The research works extends the boundaries of technology capabilities by creating a new immersive VR Property Showhouse enabling users to explore properties from the safety and convenience of their own homes, guaranteeing both their well-being and ease. While numerous existing implementations have predominantly focused on websites, AR, immersive and non-immersive VR technologies, this research contributes by investigating the user acceptance and usability of immersive VR technology specifically for viewing property showhouses among young adults aged between 18 and 24 years. The findings bear witness to the resounding success of the application in engendering an immersive and captivating virtual environment that effectively evokes a profound sense of presence within the showcased property. Notably, the personalized feature allowing modification of wall colors garnered positive feedback, empowering participants to envision diverse aesthetic possibilities and customize properties according to their personal preferences. Moreover, the VR Property Showhouse exhibited a good level of usability, as indicated by the highly favorable scores obtained through the SUS evaluation, affirming that most participants found the application to be intuitive and easy to use. However, it is important to acknowledge the limitations of this study. The sample size was relatively small, primarily comprising of young adults aged 18 to 24 years, thereby impeding the generalizability of the findings to a broader population. Future research should encompass a more diverse participant pool, spanning various demographic groups, to ensure a comprehensive understanding of user experiences across diverse segments of society. Looking ahead, future directions for the VR Property Showhouse could incorporate additional interactive features, such as virtual furniture arrangement, to provide users with a more comprehensive comprehension of the property's layout and spatial relationships. Furthermore, an exploration of the long-term effects of utilizing VR technology for property viewing, including its influence on decision-making processes and subsequent purchase intentions, holds considerable value for both researchers and industry practitioners. In summary, the VR Property Showhouse evaluation revealed its potential as an immersive and captivating tool for property viewing. The findings add to existing knowledge in the field of VR while emphasizing the importance of user-centric design and personalization in creating truly captivating virtual experiences. The immersive VR Property Showhouse has the potential to revolutionize the way properties are viewed, evaluated, and experienced by addressing its limitations and pursuing future directions.

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